

Mohammed Maalin

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EDUCATION

The Ohio State University

Columbus, OH

B.S in Computer Science & Engineering | Specialization: Artificial Intelligence

GPA: 3.6 | May 2026

Honors: Dean's List, Trustees Scholar, Choose Ohio First Scholar (1 of 80)

Coursework: Operating Systems, Networking, Programming Languages, Machine Learning, Neural Networks

Clubs: Claude Builders Club, OSU x 99P Labs AI Reading Club (Discussion lead), ColorStack

TECHNICAL SKILLS

Programming Languages: Rust, Go, Python, TypeScript

Infrastructure: Docker, Kubernetes, OpenShift, Linux

Frameworks: Model Context Protocol, FastAPI, Next.js, Rails, LangChain

Data & Cloud: MongoDB, SQLite, Azure, Firebase, AWS, Convex

WORK EXPERIENCE

Wells Fargo

Columbus, OH

Software Engineer Intern (Platform + AI)

Jun – Aug 2025

- Built a triage agent that used Jira MCP tools to read stack traces and create tickets; reduced triage latencies by 90%; deployed on internal OpenShift pods
- Shipped a codegen pipeline adopted by 12 engineering teams; automated scaffolding of new Java microservices
- Maintained Rust/Docker tooling used daily by platform engineering team, owned technical documentation

Ohio State College of Medicine

Columbus, OH

Undergraduate Research Assistant

Apr – Dec 2025

- Built web scrapers with caches to automate profile collection across all major US residencies and medical schools
- Built an ETL pipeline processing 10K+ generated profiles into validated, structured datasets using Python
- Applied NLP methods to assess overlooked topics in first year medical school curricula

PROJECTS

Neural Networks SER competition (2nd place) | PyTorch, Torchaudio, NumPy

- Built speech emotion recognition model on limited audio dataset using mel spectrogram features; achieved 70% accuracy and placed 2nd overall
- Improved generalization using SpecAugment, reduced overfitting by switching LSTM to GRU layers with dropout regularization

Goldfish, Capstone project | FastAPI, Convex, OpenRouter, Exa MCP

- Integrated HippoRAG knowledge-graph memory system to share contextual memory across different LLM providers
- Implemented streaming web-search pipeline using Exa MCP to augment responses with real-time context
- Added Convex-backed persistence layer for long-term memory, user settings, and agent state management

PropheSea, HackPrinceton submission (Best Predictive Intelligence) | React, MCP, Next.js, FastAPI, Grok

- Built agentic trading platform for Kalshi markets; used Grok search to gather real-time sentiments
- Implemented event-driven pipeline to process implied volatility and generate real-time bet recommendations
- Ran historical backtests over NOAA data; achieved 82% accuracy

Aurora Search API | FastAPI, AWS EC2, Docker

- Deployed message search API on EC2 with in-memory store and message prefetching; <10ms responses

Adobe Flash clone | Next.js, HTML5 Canvas API, Vercel

- Built browser-based frame-by-frame animation tool; 100+ users
- Implemented various brush sizes, adjustable frame-rate, onion skin for better user experience